

CROWDBEATS V2 — PUBLIC DISCOVERY, NEARBY MUSIC MAP & LOCATION SEARCH IMPLEMENTATION

You are modifying the CURRENT Crowdbeats V2 mobile + cross-platform web application.

This is an ADDITIVE implementation.

DO NOT rebuild the application from scratch.

DO NOT replace the existing design system.

DO NOT remove existing Stripe Connect, Firebase, moderation, compliance, authentication, role, payment, creator, band, campaign, admin, or security functionality.

Use the existing Crowdbeats architecture and existing Google Stitch-derived UI/UX.

The attached/reference Crowdbeats "Nearby Live Music" design should guide the experience.

The desired experience should feel as effortless as Uber, Lyft, Google Maps, Spotify discovery, and modern location-based mobile applications while remaining visually unique to Crowdbeats.

PRIMARY PRODUCT REQUIREMENT

When ANY person opens Crowdbeats, including:

- a visitor who does not have an account
- a logged-out fan
- a logged-out solo musician
- a logged-out band member
- an existing logged-in user

they must be able to browse Crowdbeats WITHOUT authentication.

Authentication is NOT required merely to discover music.

Public users must be able to:

view Crowdbeats Home

view Nearby

view the musician/band map

browse musicians and bands

view public musician profiles

view public band profiles

view public performances

view public venues

view public campaigns where permitted

search other cities/towns/states/countries

switch between map/list views

browse categories

view popular artists

view trending artists

view musicians nearby

view bands nearby

view live performers

view upcoming performers

view genres

view venues

view profile media approved for public display

Authentication must occur only when the visitor attempts an authenticated action.

AUTHENTICATION GATING MODEL

DO NOT put an authentication wall in front of public discovery.

Use progressive authentication.

PUBLIC ACTIONS:

Home browsing

Nearby browsing

Map browsing

Search

Viewing artist profiles

Viewing band profiles

Viewing venue pages

Viewing public campaigns

Viewing public photos/media

Viewing genres

Viewing approximate distance

Viewing public performance information

AUTHENTICATION REQUIRED:

Tip

Support financially

Save payment method

Follow

Favorite

Message creator

Comment

Report where identity is required by policy

Create campaign

Create musician profile

Create band

Edit profile

Access personal activity

Access transaction history

Access payouts

Access Stripe Connect

Access account settings

Sponsor financial actions

Admin actions

CRITICAL AUTHENTICATION UX

When an unauthenticated visitor taps:

TIP

do NOT simply redirect them to a generic login page and lose their current context.

Persist the intended action.

Example:

Visitor views:

Jake Rios Tip \$20

Visitor taps:

Tip \$20

Crowdbeats should preserve:

creatorId creatorSlug selectedTipAmount sourceScreen performanceId if applicable venueId if applicable

Then show:

"Sign in to tip Jake Rios"

Options:

Continue with Google

Continue with Apple where supported

Sign in

Create account

After successful authentication:

return immediately to the Jake Rios tipping flow

with the originally selected \$20 amount preserved.

Do NOT automatically charge.

The authenticated user must still confirm the payment.

PHASE 0 — AUDIT EXISTING IMPLEMENTATION

Before making modifications:

Inspect the current Crowdbeats repository.

Identify existing:

Home

Nearby

Map

artist profiles

band profiles

venue profiles

Firebase structure

authentication

Stripe tipping

creator search

location services

Google Maps integration

Google Places integration

geolocation logic

Firestore indexes

App Check

Cloud Functions

security rules

navigation

deep links

design tokens

Google Stitch components

Do not recreate functionality that already exists.

Create:

docs/discovery/PUBLIC_DISCOVERY_CURRENT_STATE_AUDIT.md

Classify requirements:

IMPLEMENTED

PARTIAL

MISSING

NEEDS REFACTOR

SECURITY ISSUE

PRIVACY ISSUE

Continue with additive implementation after the audit.

PUBLIC APPLICATION ENTRY

Change the application startup flow where necessary.

The application must NOT require login immediately after splash screen.

Desired flow:

Crowdbeats Splash

↓

Public Crowdbeats Home

Users can discover music immediately.

Provide Sign In / Account entry naturally through Profile or authenticated actions.

Do not present authentication as the first mandatory screen.

PRIMARY MOBILE NAVIGATION

Preserve or implement a structure similar to:

HOME

NEARBY

TIP / QR

ACTIVITY

PROFILE

When unauthenticated:

HOME = public

NEARBY = public

TIP scanner = may open publicly but authentication is required before completing monetary transaction

ACTIVITY = authentication required

PROFILE = shows Sign In / Create Account

When the user taps a protected tab, show the authentication gate without destroying navigation state.

NEARBY SCREEN

Implement or enhance the Nearby experience shown in the Crowdbeats design.

Header:

Nearby Live Music

Location indicator below header.

Example:

San Diego, CA

Location selector must be interactive.

Provide:

Map

List

Venues

Filter

Map and List must represent the same discovery query.

LOCATION SOURCE

Crowdbeats should determine discovery location using this hierarchy:

1. user-selected searched location
2. current device location if permission granted
3. previous non-sensitive discovery location stored locally
4. reasonable default discovery experience

Never require precise location permission to use Crowdbeats.

If location permission is denied:

Crowdbeats must still work.

Display:

"Search a city or area"

and permit complete public discovery through location search.

LOCATION PERMISSION UX

Do not request location permission unnecessarily during splash.

Allow the visitor to enter Crowdbeats first.

When Nearby is opened:

explain the value before requesting permission.

Example:

"Find live music near you"

Use location

or

Search another area

If declined:

do not repeatedly nag the user.

LOCATION SEARCH

Add a prominent search control to Nearby and discovery.

Placeholder:

Search city, town, state or country

Examples:

Torrance, California

Palm Springs, California

Los Angeles, California

Austin, Texas

Nashville, Tennessee

London, United Kingdom

Auckland, New Zealand

Use the supported Google Maps Platform / Places autocomplete technology already compatible with the project.

Autocomplete must provide geographic suggestions while typing.

Search should prioritize geographic entities such as:

city

town

locality

administrative area

state/province/region

country

Do not populate artist names into this particular LOCATION search field.

Artist/music search may exist separately.

AUTOCOMPLETE UX

Example:

User types:

Tor

Crowdbeats may display:

Torrance, CA, USA

Toronto, ON, Canada

Torquay, England, UK

Each result should include enough geographic context to distinguish locations.

When the visitor selects:

Torrance, CA, USA

Crowdbeats should:

resolve the selected geographic place

store the appropriate safe location identifiers

center the discovery map

update nearby query

update heading to:

Torrance, CA

refresh categories

refresh musician results

refresh band results

refresh live performances

refresh venues

Do not reload the entire application unnecessarily.

LOCATION SEARCH DATA MODEL

Where supported, store safe discovery information such as:

placeId

displayName

city/locality

administrativeArea

country

latitude

longitude

viewport/bounds

Do not treat formatted location strings as primary identity when a stable provider identifier is available.

Do not store a visitor's location history server-side without an explicit product need and privacy review.

Public visitors' selected search areas should preferably remain local/session-based.

MAP EXPERIENCE

Implement a premium interactive music discovery map.

Support:

artist pins

band pins

live performance pins

venue pins where appropriate

clustered results when dense

selected marker state

current search-area center

optional device-location indicator

selected artist preview

Map marker interaction:

tap marker

↓

preview musician/band

↓

show:

artist image

artist/band name

verified badge if legitimately verified

distance

genre

venue

live indicator

watching/viewing information only if legitimate data exists

Tip button

View Profile

MAP PRIVACY REQUIREMENT

NEVER publicly expose a musician's private residential address or continuously tracked precise location.

Permitted public location sources should include:

verified venue

scheduled performance location

artist-approved public performance position

approved general discovery area

public venue coordinates

For a creator who is not actively performing:

use city/general discovery association where appropriate.

Do not place a pin on their home.

Do not infer home location from billing information, Stripe information, IP address, Firebase profile data, or private geolocation history.

LIVE LOCATION MODEL

If Crowdbeats supports "Live Now":

a creator must explicitly activate or confirm public live-performance status.

Where applicable validate:

creator eligibility

venue/performance association

time window

public visibility

moderation eligibility

location-sharing authorization

Never silently broadcast a creator's background GPS position.

MAP CATEGORIES

Create useful discovery categories.

Core categories should include:

LIVE NOW

NEAR YOU

POPULAR

TRENDING

SOLO MUSICIANS

BANDS

VENUES

UPCOMING

GENRES

FEATURED

NEW TO CROWDBEATS

You may add relevant categories only when backed by real application data.

Do not manufacture popularity metrics.

HOME DISCOVERY

The public Crowdbeats home should immediately demonstrate value.

Potential sections:

Live Near You

Popular Near You

Trending Musicians

Bands Near You

Tonight

Upcoming This Week

Featured Artists

Popular Genres

Venues Near You

New on Crowdbeats

Campaigns Worth Discovering where policy permits

Recommendations should degrade gracefully for visitors without accounts.

Do not require personalization data for public discovery.

NEARBY LIST VIEW

Create synchronized Map and List experiences.

List cards should include where available:

creator photo

band photo

name

verified state

distance

genre

live status

venue

performance time

favorite control

Tip button

Public users may see the Tip button.

Tapping Tip while logged out invokes authentication.

Do NOT hide the existence of tipping from public visitors.

VENUES VIEW

Nearby → Venues

Display public venues with active or upcoming Crowdbeats musicians.

Venue results can include:

venue name

distance

performers tonight

number of active musicians

genres

event times

[View Venue](#)

[View Performers](#)

Do not create venue data from unverified arbitrary addresses.

SEARCH ARCHITECTURE

Crowdbeats needs TWO clearly distinct search experiences.

LOCATION SEARCH:

Search geographical area.

Examples:

Torrance, CA

San Diego, CA

Nashville, TN

ARTIST DISCOVERY SEARCH:

Search:

solo musician name

band name

genre

venue

campaign where appropriate

Do not mix them in a confusing manner.

A universal search may later combine these using clearly labeled result groups.

SEARCH AREA MODE

When a user searches another city:

Crowdbeats must clearly communicate that results are for the searched area rather than current device location.

Example:

Nearby Live Music

Torrance, CA

Provide:

Use My Location

button/action.

Do not overwrite the user's actual current GPS location with the searched location.

Maintain:

deviceLocation

separately from:

discoveryLocation

DISTANCE

Distance must be calculated from:

current device location when Nearby uses user's location

OR

the center/relevant point of the manually selected discovery area when browsing another location

Clearly avoid misleading wording.

If manually searching Torrance while physically in San Diego, do not display distance as if the visitor is physically in Torrance without appropriate context.

For searched areas, terms such as:

"within 2 mi of Torrance"

may be preferable.

FIRESTORE GEO DISCOVERY

Review the existing Firebase architecture.

Implement scalable geographic discovery.

Do NOT download the entire creator collection and filter it on the client.

Use an appropriate geographic indexing strategy compatible with Firestore.

Potential architecture may include:

geohash

latitude

longitude

publicLocationType

creatorId

performanceId

venueId

isLive

visibility

discoveryEligible

updatedAt

All query results must be filtered against server-authoritative eligibility.

Only ACTIVE and publicly discoverable creators/content may appear.

PUBLIC DISCOVERY ELIGIBILITY

Create centralized server logic or equivalent enforcement such as:

```
assertCreatorPubliclyDiscoverable()
```

A creator should appear in public discovery only when appropriate.

Possible checks:

```
profileActive == true
```

```
publicProfileEnabled == true
```

```
moderationStatus permits visibility
```

```
not suspended
```

```
not terminated
```

```
creator role valid
```

```
profile has sufficient required data
```

```
public discovery location valid
```

If Live Now:

```
live state valid
```

```
performance visibility valid
```

```
location authorization valid
```

This is separate from monetization eligibility.

A creator could potentially remain publicly visible but be demonetized.

PUBLIC PROFILE ROUTES

Solo artists:

/artist/{creatorSlug}

Bands:

/band/{bandSlug}

Public visitors must be able to open these URLs without login.

Public profile information may include:

public artist name

band name

profile image

cover image

bio

genres

public media

verified badge

public social links where allowed

upcoming performances

active campaigns

public venue information

Tip button

Do not expose:

email

phone number

private address

bank information

Stripe account information

private legal name unless creator intentionally publishes it

identity verification documents

moderation notes

internal flags

private analytics

precise private location

TIP AUTH GATE

This requirement is mandatory.

Unauthenticated visitors CAN:

see Tip

choose artist

open tip interface

optionally select amount before login

but cannot create or confirm a Stripe payment until authenticated.

Example:

Tap Tip

↓

\$5 \$10 \$20 Custom

↓

Visitor selects \$20

↓

Authentication required

"Sign in to tip Jake Rios"

↓

Authenticate

↓

return to:

Jake Rios \$20 selected

↓

review payment

↓

Stripe PaymentIntent

↓

confirm

Do not automatically execute payment immediately after authentication.

STRIPE CONNECT REQUIREMENT

All creator tipping must continue to respect Crowdbeats Stripe compliance architecture.

Authentication does NOT automatically mean the creator can receive payment.

Backend must verify:

fan authenticated

creator exists

creator public identity valid

creator monetization ACTIVE

Stripe Connected Account valid

charges enabled

creator not demonetized

creator not suspended

creator not terminated

no compliance restriction preventing payment

tip amount valid

currency valid

payment request idempotent

App Check valid where supported

If any requirement fails:

payment must fail closed.

FAVORITES / FOLLOW

A public visitor may tap a heart/follow icon.

If unauthenticated:

show:

"Sign in to follow Jake Rios"

After authentication:

return to the artist and complete the requested follow action with user confirmation where appropriate.

Do not silently create unnecessary accounts.

DEEP LINK CONTINUITY

Preserve public deep links across authentication.

Example:

crowdbeats.ai/artist/jake-rios

Visitor taps Tip

authenticates

returns to Jake Rios.

Do not redirect authenticated users to a generic dashboard after login when they arrived from a specific creator action.

WEB CROSS-PLATFORM REQUIREMENT

The same discovery architecture must work on Crowdbeats web.

Desktop/tablet web visitors must be able to:

browse without login

search locations

use Map/List

view artists

view bands

view venues

view campaigns where public

open creator URLs

start Tip flow

Authentication is required before completing protected actions.

Ensure responsive implementation.

Do not create separate incompatible discovery models for mobile and web.

Use shared schemas/business logic where appropriate.

GOOGLE MAPS SECURITY

Do not expose unrestricted API credentials.

Restrict Google Maps Platform credentials appropriately by:

application

platform

domain

bundle/package identifier

allowed APIs

Use server-side services for operations that require protected server credentials.

Do not commit secrets.

FIREBASE SECURITY RULES

Public users should have read access ONLY to intentionally public discovery data.

Never make entire creator documents public just because profiles are public.

Prefer public-safe projections/documents or strict field architecture where appropriate.

Unauthenticated users must NEVER be permitted to:

write artist profiles

alter bands

change locations

create Stripe records

alter payment state

modify moderation state

change verification

modify popularity metrics

create fake live sessions

change creator eligibility

write administrative data

Public discovery must be read-only.

PUBLIC DATA PROJECTION

Evaluate separating sensitive creator data from public profile data.

Example concept:

users/{uid}

private account information

creatorPrivate/{creatorId}

sensitive creator/application information

creatorPublic/{creatorId}

safe discovery information

Do not migrate recklessly.

Adapt this model to the canonical existing Crowdbeats architecture.

POPULARITY & TRENDING

Do not let clients directly write:

popularityScore

trendingScore

viewCount

tipCount

rankingScore

Calculate sensitive ranking inputs through trusted backend processes.

Prevent creators from gaming rankings by client writes.

Potential ranking signals may include legitimate:

profile engagement

unique views

performance activity

favorites

recent activity

verified live events

aggregated tipping signals where privacy-safe

Do not expose individual fan financial data.

CATEGORY QUERY DESIGN

Provide data-driven queries for:

Nearby Live

Nearby Artists

Nearby Bands

Popular Near Area

Trending Near Area

Upcoming Near Area

Featured

Genre

Venues

Results should support:

pagination

loading state

empty state

error state

offline/retry state where appropriate

Do not fetch unlimited collections.

EMPTY STATES

Example when no musicians are nearby:

"No live music found nearby."

Options:

Expand area

Search another location

See upcoming performers

Explore popular musicians

Never show fake musicians simply to populate UI.

FILTERS

Nearby filtering may include:

Live Now

Solo

Bands

Venue

Distance radius

Genre

Tonight

This Week

Verified

Use clear filters.

Do not make verified status a requirement for all discovery unless business policy requires it.

PERFORMANCE

Map browsing must remain smooth.

Implement:

pagination

query caching

marker clustering

debouncing

autocomplete request management

image optimization

lazy loading

reasonable geographic query limits

Do not run a Firestore query for every keystroke unnecessarily.

Debounce text input.

Use provider session/token practices where applicable.

ACCESSIBILITY

Support:

screen readers

semantic labels

keyboard navigation on web

sufficient contrast

accessible map/list alternative

large touch targets

dynamic text

Do not make map-only discovery the sole means of finding musicians.

List View must remain fully functional.

ANALYTICS PRIVACY

Allow privacy-reviewed aggregated discovery events such as:

nearby_opened

discovery_location_selected

artist_profile_viewed

band_profile_viewed

map_marker_selected

location_search_completed

tip_auth_gate_opened

Do NOT send:

precise continuous GPS history

payment card information

private messages

identity documents

raw authentication tokens

private creator addresses

to analytics.

TESTING REQUIREMENTS

Create automated tests proving:

public visitor can open application without login

public visitor can access Home

public visitor can access Nearby

public visitor can search Torrance, California

location autocomplete returns/selects valid place information using mock/test provider behavior

public visitor can switch Map/List

public visitor can open artist profile

public visitor can open band profile

public visitor cannot modify artist data

public visitor cannot access private creator fields

public visitor cannot directly create Stripe payment

public visitor tapping Tip receives authentication gate

selected tip context survives authentication

authentication returns visitor to creator

payment still requires explicit confirmation

demonetized creator cannot receive tip

suspended creator cannot receive tip

terminated creator excluded from discovery where required

private residential location never appears in public discovery payload

manual searched area remains separate from device GPS location

creator cannot manipulate ranking score client-side

creator cannot fake verified live location client-side

unauthorized location writes are rejected

Google/Firebase secrets are not exposed

mobile and web use compatible discovery data

E2E TEST SCENARIO 1 — PUBLIC TORRANCE DISCOVERY

Create deterministic test:

launch Crowdbeats logged out

do not display mandatory authentication

open Nearby

tap location search

enter:

Torrance

select:

Torrance, CA, USA

map centers on Torrance discovery area

nearby results update

select musician

open musician public profile

tap Tip

choose \$20

Crowdbeats requests authentication

authenticate with test fan account

return to same musician

preserve \$20

show payment confirmation

complete Stripe TEST MODE payment

verify transaction result

verify no duplicate payment

E2E TEST SCENARIO 2 — LOCATION DENIED

launch application

deny device location permission

open Nearby

application remains functional

search:

Nashville, Tennessee

select Nashville

display discovery results

open band profile

verify no authentication required

E2E TEST SCENARIO 3 — PRIVACY

create test musician with:

private home address

private precise coordinates

public city

public scheduled venue

search nearby

verify:

home address NEVER returned

private coordinates NEVER returned

public venue/performance location may appear

DESIGN REQUIREMENT

Preserve Crowdbeats premium dark visual language.

Use the existing Google Stitch design system.

The reference design includes:

black/dark background

premium purple accent

interactive artist map markers

Map/List/Venues segmented navigation

floating location controls

artist cards

Live indicators

distance

genre

venue

watching/engagement metadata

Tip CTA

bottom navigation

Do not blindly reproduce every placeholder value in the design.

Use live application data.

No fake production metrics.

Do not redesign unrelated Crowdbeats screens.

DESIRED NEARBY SCREEN

Approximate hierarchy:

Nearby Live Music

Current/Search Location

[Search city, town, state or country]

Map | List | Venues

Filters

Interactive map

map markers

selected marker preview

Live Near You

artist/band cards

Tip

Bottom Navigation

SEARCH EXAMPLE

User:

Torrance

Autocomplete:

Torrance, CA, USA

User selects result.

Header:

Nearby Live Music Torrance, CA

Categories:

Live Now Popular Solo Artists Bands Venues Upcoming

Results should now represent the selected Torrance discovery area.

Provide:

Use My Location

to return to local discovery.

CRITICAL SECURITY REQUIREMENT

PUBLIC BROWSING DOES NOT MEAN PUBLIC DATABASE ACCESS.

Expose only fields intentionally designed for public consumption.

Continue enforcing private fields through backend architecture and Firebase Security Rules.

CRITICAL STRIPE REQUIREMENT

Do not weaken Stripe compliance because browsing is public.

Public visitors may discover creators.

ONLY authenticated customers may initiate protected payment operations.

ONLY eligible Stripe Connect creators may receive payments.

Stripe remains the financial system of record.

CRITICAL PRIVACY REQUIREMENT

Never expose someone's private/home address to make Nearby work.

Never continuously broadcast musicians' GPS locations.

"Nearby" means discovery based on approved public performance/discovery location.

IMPLEMENTATION DELIVERABLES

After implementation create:

docs/discovery/PUBLIC_DISCOVERY_CURRENT_STATE_AUDIT.md

docs/discovery/PUBLIC_DISCOVERY_ARCHITECTURE.md

docs/discovery/GEO_SEARCH_ARCHITECTURE.md

docs/discovery/PUBLIC_PROFILE_SECURITY_MODEL.md

docs/discovery/TIP_AUTHENTICATION_GATE.md

docs/discovery/DISCOVERY_PRIVACY_REVIEW.md

docs/discovery/PUBLIC_DISCOVERY_TEST_REPORT.md

Update Stripe compliance documentation to accurately state:

Public Crowdbeats creator/discovery content can be browsed without an account.

Authentication is required for protected actions including tipping.

Do not claim every user must log in to browse.

FINAL REPORT

When complete, return:

1. Existing components reused
2. Components created
3. Components modified

4. Routes created
5. Firestore collections/documents modified
6. Firestore indexes created
7. Firebase Security Rules modified
8. Cloud Functions created/modified
9. Google Maps/Places integration changes
10. Authentication gating changes
11. Stripe tipping changes
12. Public/private creator data separation
13. Location privacy protections
14. Map implementation
15. Search implementation
16. Category implementation
17. Mobile implementation
18. Web implementation
19. Tests passed
20. Tests failed
21. Remaining security concerns
22. Remaining privacy concerns
23. Remaining Stripe compliance concerns

Explicitly tell me whether these requirements are now TRUE:

PUBLIC_DISCOVERY_WITHOUT_AUTH = TRUE/FALSE

PUBLIC_ARTIST_PROFILES_WITHOUT_AUTH = TRUE/FALSE

PUBLIC_BAND_PROFILES_WITHOUT_AUTH = TRUE/FALSE

LOCATION_AUTOCOMPLETE = TRUE/FALSE

SEARCH_OTHER_CITIES = TRUE/FALSE

MAP_DISCOVERY = TRUE/FALSE

MAP_LIST_SYNC = TRUE/FALSE

LOCATION_PERMISSION_OPTIONAL = TRUE/FALSE

PRIVATE_HOME_LOCATION_PROTECTED = TRUE/FALSE

TIP_REQUIRES_AUTH = TRUE/FALSE

TIP_CONTEXT_PRESERVED_THROUGH_AUTH = TRUE/FALSE

STRIPE_CONNECT_ELIGIBILITY_ENFORCED = TRUE/FALSE

FIREBASE_PUBLIC_DATA_ISOLATION = TRUE/FALSE

MOBILE_TESTS_PASS = TRUE/FALSE

WEB_TESTS_PASS = TRUE/FALSE

Do NOT mark TRUE without actual implementation and passing evidence.

DO NOT deploy to production.

Stop after implementation/testing/report generation and wait for authorization.